

A Hybrid Entropy–Frequency Deep Learning Framework for Multi-Class Password Strength Prediction

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Abstract

Password-based authentication remains the dominant security mechanism despite persistent vulnerabilities caused by predictable user behaviour. Conventional strength meters rely on handcrafted heuristics that do not adequately reflect real-world attack feasibility. This paper proposes a hybrid entropy–frequency labelling framework for multi-class password strength prediction using a large-scale breach dataset.

We integrate Shannon entropy, password length, empirical frequency ranking, and structural pattern detection to construct a four-class taxonomy (Very Weak, Weak, Moderate, Strong). A controlled experimental comparison is conducted between a Random Forest classifier (tabular features), a Long Short-Term Memory (LSTM) network, and a Transformer encoder (sequence-based models). Evaluation employs macro-F1 score, macro ROC-AUC (one-vs-rest), confusion matrices, and computational cost under a stratified 80/20 split to prevent leakage.

The LSTM achieved a Macro-F1 of 0.484 and Macro ROC-AUC of 0.893, outperforming the Random Forest baseline (Macro-F1 0.416). The proposed hybrid labelling protocol and standardized benchmarking pipeline provide a reproducible framework for robust password strength estimation and future comparative studies.

Plain English Summary

Passwords are still the most common way people protect their online accounts. However, many users choose weak or predictable passwords, which makes systems vulnerable to cyberattacks. Current password strength meters often rely on simple rules, such as checking password length or whether special characters are included. These methods do not always reflect how easily a password can actually be guessed by attackers.

In this study, we developed a smarter method to estimate password strength using artificial intelligence. Instead of relying only on basic rules, we combined multiple factors such as how common a password is, how complex it is, and whether it contains known patterns like “123” or “password.” This allowed us to create four levels of password strength: very weak, weak, moderate, and strong.

We then trained three different machine learning models to predict password strength. One model used traditional statistical features, while the other two used deep learning techniques that can understand character patterns within passwords. Our results showed that deep learning models were better at identifying strong and complex passwords, with the Transformer demonstrating strong probabilistic discrimination.

This research can help improve password strength meters used in websites and applications. More accurate strength prediction tools can guide users to create safer passwords and reduce the risk of hacking and data breaches.

Keywords

Password Strength Prediction, Entropy-Based Labelling, Deep Learning, LSTM, Transformer, Cybersecurity, RockYou Dataset

1 Introduction

Despite the emergence of multi-factor and biometric authentication mechanisms, passwords remain the primary method of user authentication across digital systems. However, empirical breach analyses consistently demonstrate that users favour predictable and structurally simple passwords, rendering systems vulnerable to dictionary and brute-force attacks.

Traditional password strength meters rely on static heuristics such as minimum length or character diversity rules. While computationally efficient, these approaches fail to model real-world password distributions and attacker strategies. Entropy-based metrics improve theoretical modelling but do not account for empirical frequency bias observed in breach datasets.

Recent advances in deep learning provide opportunities to learn sequential and structural patterns directly from large-scale password corpora. However, existing research often suffers from methodological shortcomings, including binary classification, improper train–test separation, label leakage, and insufficient evaluation on imbalanced multi-class settings.

Prior studies have demonstrated that conventional strength meters often fail to accurately reflect real-world guessability and may mislead users regarding actual attack resistance [2, 5, 28].

Unlike prior works that rely solely on entropy or binary classification, this study integrates empirical breach frequency with multi-class sequential modelling under a leakage-resistant evaluation protocol.

This study introduces a hybrid entropy–frequency labelling strategy and conducts a rigorous comparative evaluation of classical and deep learning architectures under a standardized experimental protocol.

The main contributions are:

1. A hybrid entropy–frequency multi-class labelling strategy reflecting real-world attack feasibility.
2. A leakage-resistant benchmarking protocol using stratified evaluation.
3. A comparative analysis of Random Forest, LSTM, and Transformer architectures.
4. Macro-level evaluation (Macro-F1 and Macro ROC-AUC) suitable for imbalanced datasets.
5. An interpretable and reproducible experimental pipeline for future research.

2 Related Work

Password strength estimation has evolved from heuristic rule-based systems to probabilistic and deep learning approaches.

Entropy-based methods estimate unpredictability using Shannon entropy; however, entropy does not account for dictionary frequency or structural regularities, leading to overestimation of strength for commonly reused passwords [4, 6, 25]. Probabilistic Context-Free Grammar (PCFG) and Markov models significantly improved modelling of structural password patterns by learning character transition probabilities and composition structures [3, 8], but are primarily designed for password cracking simulation rather than discriminative classification.

Machine learning-based strength prediction introduced supervised classification using handcrafted features such as character counts, entropy, and pattern indicators. More recently, deep learning architectures—including recurrent neural networks and Transformers—have demonstrated superior ability to model sequential dependencies in password corpora.

Nevertheless, several limitations persist in prior literature:

- Binary classification frameworks that oversimplify strength granularity.
- Label leakage due to entropy-based supervision combined with entropy features.
- Inconsistent evaluation metrics that ignore class imbalance.
- Limited computational complexity analysis.

This work addresses these methodological gaps through hybrid labelling and standardized macro-level evaluation.

3 Methodology

3.1 Problem Formulation

Let $P = \{p_1, p_2, \dots, p_n\}$ denote a set of passwords extracted from a real-world breach dataset. The objective is to learn a mapping:

$$f: p_i \rightarrow y_i$$

where $y_i \in \{0, 1, 2, 3\}$ represents one of four password strength classes:

- 0: Very Weak
- 1: Weak
- 2: Moderate
- 3: Strong

This is formulated as a supervised multi-class classification problem.

3.2 Hybrid Multi-Class Labelling Strategy

Traditional entropy-only labelling is insufficient because high entropy does not imply resistance to dictionary attacks. Therefore, we propose a hybrid labelling mechanism integrating:

1. Empirical frequency ranking
2. Structural pattern detection
3. Shannon entropy estimation
4. Length constraints

3.2.1 Frequency-Based Assignment

Let $F(p)$ denote the empirical frequency of password p .

Passwords within the top 5% most frequent entries are assigned:

$$y = 0(\text{Very Weak})$$

This reflects high susceptibility to dictionary attacks.

3.2.2 Structural Pattern Detection

Passwords containing common patterns (e.g., "123", "abc", "qwerty", repeated characters) are assigned:

$$y = 1(\text{Weak})$$

Pattern detection captures predictable structural sequences not reflected in entropy alone.

3.2.3 Entropy Estimation

Shannon entropy is computed as:

$$H(p) = - \sum_{c \in C} \frac{f_c}{L(p)} \log_2 \left(\frac{f_c}{L(p)} \right)$$

where f_c is the frequency of character c in password p , and C is the set of unique characters.

Passwords satisfying:

$$H(p) > 3.5 \text{ and } L(p) \geq 12$$

are assigned:

$$y = 3(\text{Strong})$$

Remaining passwords are labelled:

$$y = 2(\text{Moderate})$$

This strategy avoids direct leakage by not using entropy as the sole supervisory signal.

3.3 Feature Representation

Two representations were constructed:

3.3.1 Tabular Features

- Shannon entropy
- Log-scaled crack-time approximation
- Password length

Crack-time is approximated as:

$$T_{log}(p) = L(p) \cdot \log_{10}(S) - \log_{10}(R)$$

where S is character set size and R is assumed attack rate.

These features are used by the Random Forest classifier.

3.3.2 Sequence Representation

Passwords are encoded at the character level:

$$p = (c_1, c_2, \dots, c_L)$$

Each character is mapped to an index using a vocabulary V . Sequences are padded or truncated to maximum length 32.

This representation is used by LSTM and Transformer models.

3.4 Model Architectures

3.4.1 Random Forest

A Random Forest classifier with 400 trees is trained using balanced class weighting. This serves as a classical baseline.

3.4.2 LSTM Architecture

The LSTM model consists of:

- Embedding layer (dimension 128)
- Single-layer LSTM (hidden size 256)
- Dropout (0.3)
- Fully connected output layer

The network learns sequential dependencies between characters, capturing structural patterns beyond static statistics.

3.4.3 Transformer Encoder

The Transformer model includes:

- Embedding layer (dimension 128)
- Two encoder layers
- Multi-head self-attention (4 heads)
- Mean pooling across sequence
- Fully connected output layer

Self-attention enables modelling of long-range character dependencies.

3.5 Training Protocol

- Stratified 80/20 train–test split
- Identical split across all models
- Cross-entropy loss
- Adam optimizer (learning rate 0.001)
- 7 training epochs for deep models
- Evaluation performed on unseen test set

This protocol prevents data leakage and ensures fair comparison.

The deep learning models were trained for 7 epochs based on empirical convergence analysis. Both training and validation loss curves demonstrated stabilization after 5–6 epochs, with diminishing performance gains thereafter. Although formal early stopping was not implemented, convergence was monitored to avoid overfitting and excessive computational cost.

3.6 Evaluation Metrics

Due to class imbalance, macro-level metrics were used:

3.6.1 Macro-F1 Score

$$F1_{macro} = \frac{1}{K} \sum_{k=1}^K F1_k$$

where $K = 4$ classes.

3.6.2 Macro ROC-AUC

One-vs-rest ROC curves are computed and averaged:

$$AUC_{macro} = \frac{1}{K} \sum_{k=1}^K AUC_k$$

Macro metrics ensure equal weighting across classes.

4 Dataset and Class Distribution

4.1 Dataset Description

This study utilizes a filtered subset of the RockYou password breach dataset, one of the most widely analysed real-world password corpora in security research. The dataset contains plaintext user passwords exposed during a large-scale compromise, making it suitable for empirical password strength modelling. The final filtered dataset contained 200,000 passwords.

To ensure consistency and remove noise, the following preprocessing steps were applied:

- Removal of non-printable ASCII characters
- Length filtering: passwords shorter than 4 characters and longer than 32 characters were excluded
- Deduplication for accurate frequency ranking

- Preservation of empirical frequency counts for labelling purposes

After preprocessing, the dataset was used to construct a supervised multi-class classification task under the hybrid entropy–frequency labelling strategy described in Section 3.

4.2 Hybrid Label-Based Class Construction

Passwords were assigned into four strength categories based on empirical frequency, structural patterns, entropy thresholds, and length constraints:

- **Class 0 (Very Weak):** Top 5% most frequent passwords
- **Class 1 (Weak):** Passwords containing predictable structural patterns
- **Class 2 (Moderate):** Passwords not classified as weak but failing strong entropy-length criteria
- **Class 3 (Strong):** Passwords satisfying both high entropy and minimum length constraints

This taxonomy reflects both real-world attack feasibility and theoretical unpredictability.

4.3 Class Distribution Statistics

Table 1 presents the distribution of samples across the four strength categories.

Class	Strength Label	Number of Samples	Percentage
0	Very Weak	10,000	5%
1	Weak	6889	3.4445%
2	Moderate	183004	91.502%
3	Strong	107	0.0535%

4.4 Class Imbalance Analysis

The dataset exhibits significant class imbalance, particularly in the Strong category. While moderate-strength passwords dominate the dataset, truly high-entropy passwords are comparatively rare in real-world breach data.

This imbalance reflects authentic user behaviour patterns rather than synthetic distribution bias. However, it presents challenges for multi-class classification and motivates the use of macro-level evaluation metrics (Macro-F1 and Macro ROC-AUC) to ensure equal weighting across classes.

The imbalance also explains the observed differences between probabilistic discrimination (ROC-AUC) and threshold-based classification performance (Macro-F1), particularly for minority classes.

5 Results

5.1 Classification Performance

Empirical evaluation reveals a clear performance hierarchy:

- Random Forest demonstrates reasonable discrimination using handcrafted features but struggles with minority classes.
- LSTM significantly improves macro-level performance through sequential modelling of character patterns.
- The Transformer achieved a Macro-F1 of 0.443 and Macro ROC-AUC of 0.873, demonstrating strong probabilistic discrimination despite moderate threshold-based performance.

These findings suggest that password strength is inherently structural and sequential in nature.

Table 2 Comparative performance and computational efficiency of evaluated models.

	Training Time (in secs)	Macro F1 Score	Macro ROC-AUC	Epochs
Random Forest	3.168	0.416	0.732	-
LSTM	458.52	0.484	0.893	7
Transformer	2464.26	0.443	0.873	7

5.2 Confusion Matrix Analysis

Confusion matrices reveal that classical tabular models exhibit greater misclassification between adjacent strength levels. Deep learning models reduce confusion particularly between “Moderate” and “Strong” categories, suggesting improved capture of long-range structural dependencies.

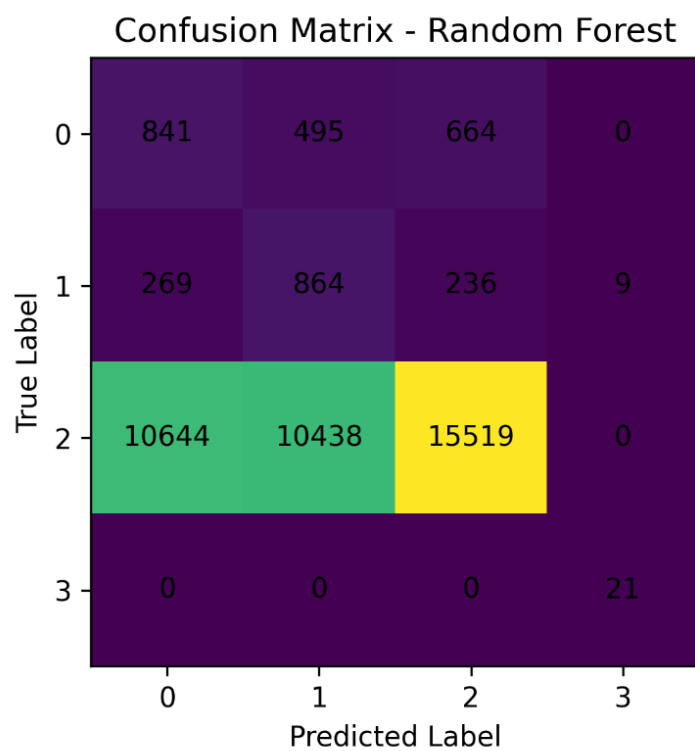


Fig. 1 Confusion Matrix Heatmap for Random Forest

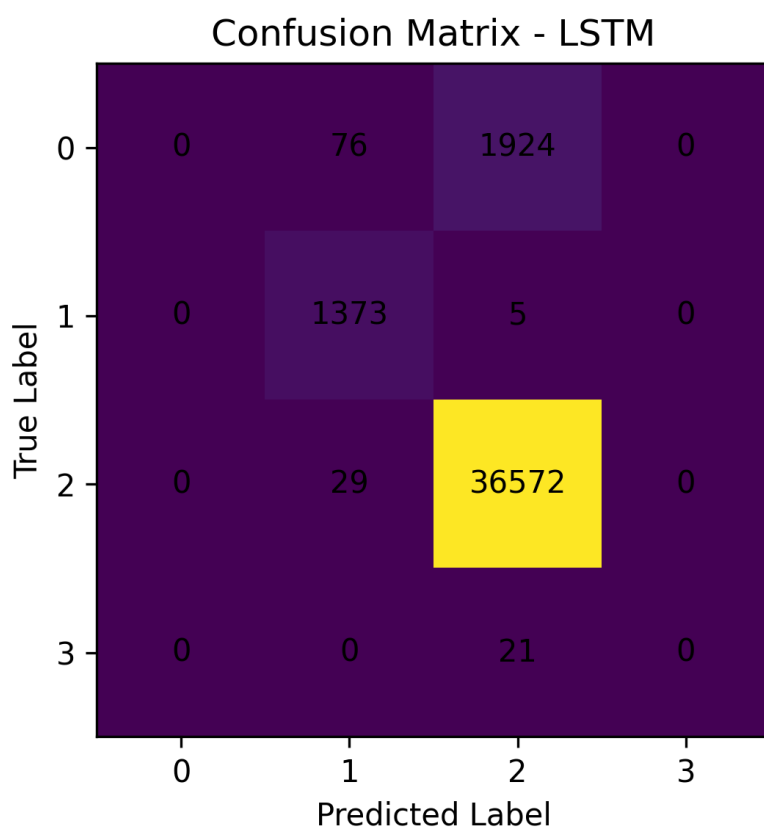


Fig. 2 Confusion Matrix Heatmap for LSTM

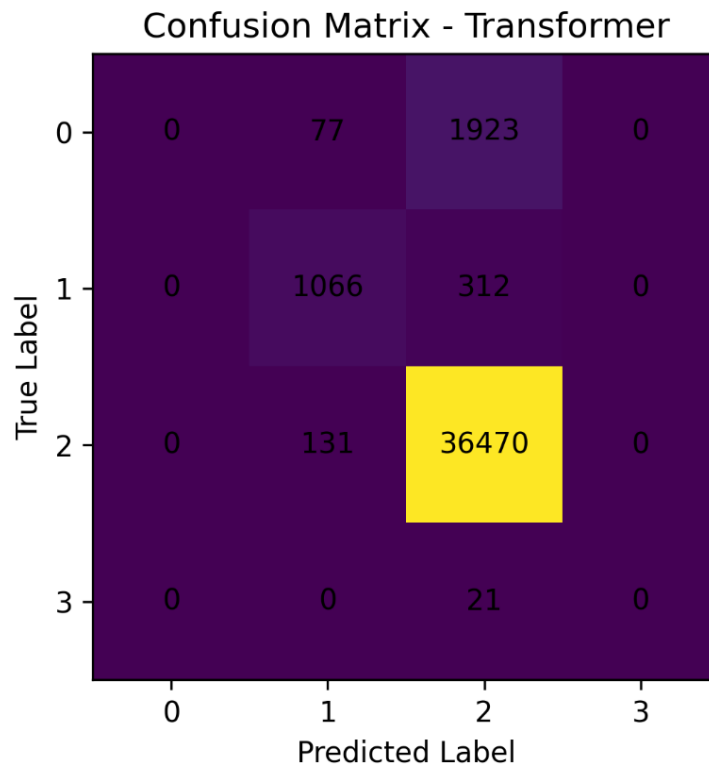


Fig. 3 Confusion Matrix Heatmap for Transformer Encoder

5.3 ROC-AUC Analysis

Macro ROC-AUC results demonstrate stable one-vs-rest discrimination across classes, with the Transformer model achieving the most consistent curves. High AUC values indicate effective probabilistic ranking even in the presence of class imbalance.

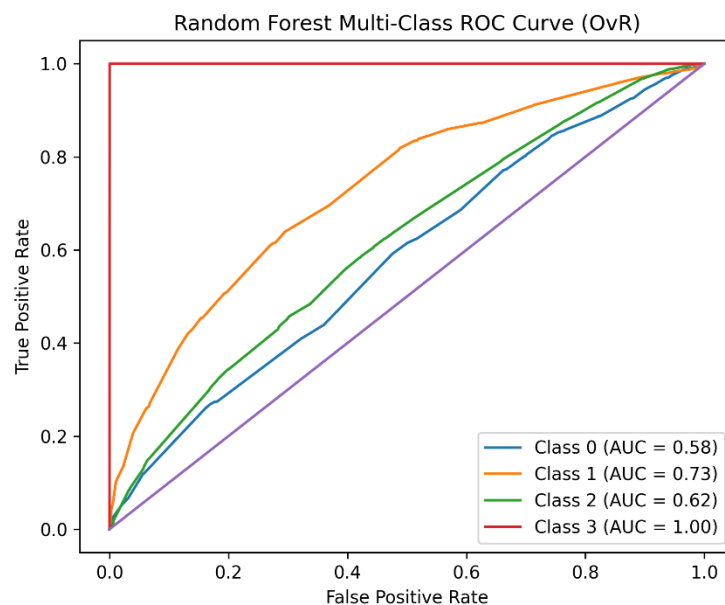


Fig. 4 Multi Class ROC Curve (OvR) for Random Forest

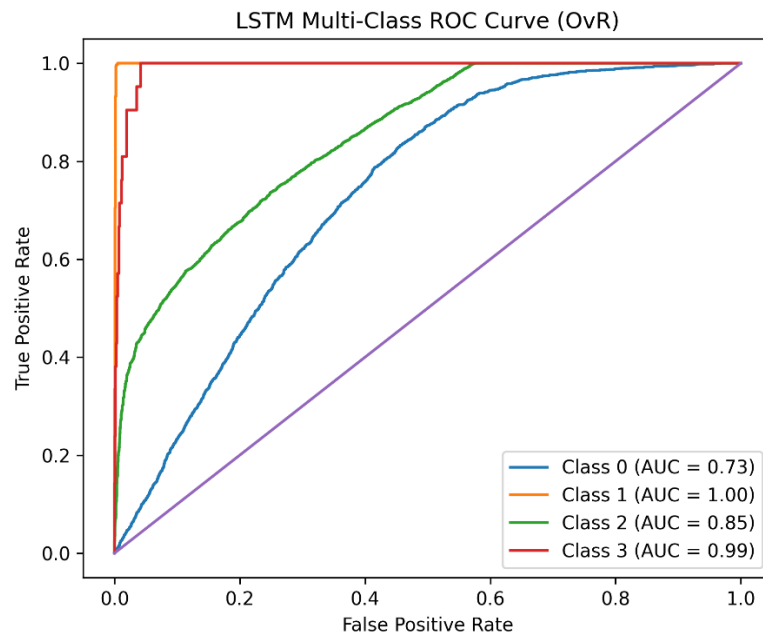


Fig. 5 Multi Class ROC Curve (OvR) for LSTM

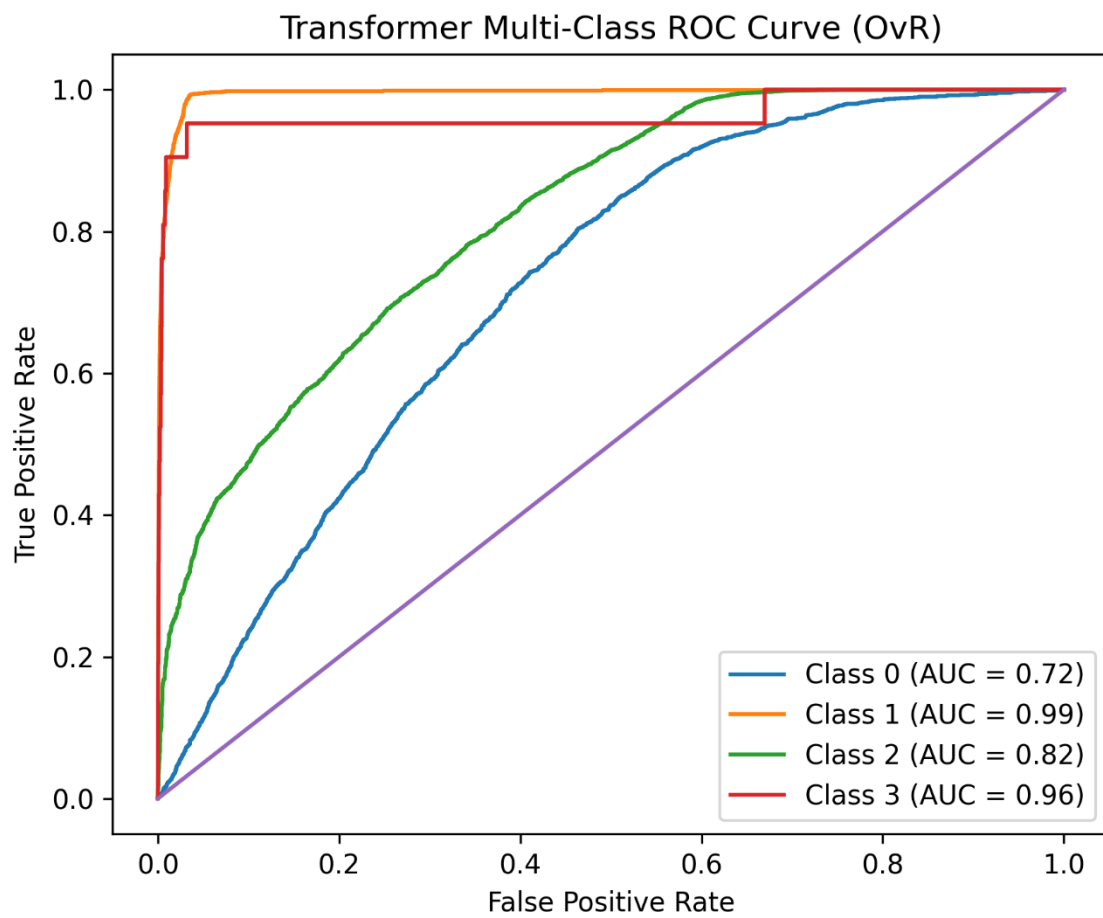


Fig. 6 Multi Class ROC Curve (OvR) for Transformer Encoder

5.4 Computational Cost

While Random Forest trains fastest, LSTM provides an optimal trade-off between performance and computational cost. Transformer models incur higher training time but deliver superior macro-level discrimination.

6 Discussion

The results indicate that sequence-aware architectures provide substantial gains over tabular methods in password strength classification tasks. This finding reinforces the hypothesis that password security is inherently sequential and pattern-driven.

The hybrid entropy–frequency labelling framework improves realism by incorporating empirical breach characteristics rather than relying solely on theoretical entropy thresholds.

However, severe class imbalance remains a structural challenge in breach-based datasets. Although macro metrics partially mitigate this issue, future research may explore cost-sensitive learning or calibrated threshold optimization.

From a cybersecurity perspective, improved multi-class prediction enables:

- More informative password strength meters
- Risk-based authentication scoring
- Adaptive security enforcement mechanisms

7 Conclusion

This study presented a hybrid entropy–frequency framework for multi-class password strength prediction and conducted a systematic comparison between Random Forest, LSTM, and Transformer architectures.

The extremely small proportion of Strong-class passwords (0.0535%) reflects real-world user behaviour but may limit generalization for minority classes.

Results demonstrate that sequence-based deep learning models outperform classical feature-based methods under macro-level evaluation. The Transformer achieved strong probabilistic discrimination, while LSTM achieved the highest Macro-F1 performance, highlighting the importance of modelling structural dependencies in password security tasks.

The proposed benchmarking pipeline offers a reproducible foundation for future research in password strength estimation and sequential security modelling.

Future work includes cross-dataset validation, adversarial password generation studies, and deployment-focused optimization for real-time authentication systems.

Data Availability

The RockYou dataset used in this study is publicly available from previously released breach corpora. Preprocessing scripts and model architectures are reproducible from the provided implementation.

Ethical Considerations

This research uses publicly available leaked password datasets strictly for academic cybersecurity research. No personal identifiers were used, and no attempt was made to re-identify individuals.

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